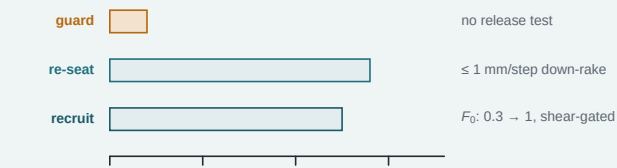
A



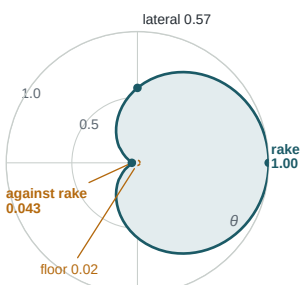
ENGAGE GATES — all four, every step

G1	press	≥ 1 sphere at $F_\perp \geq 3$ N
G2	align	$\hat{a} \cdot \hat{n}_{in} \geq 0.25$
G3	seat	spine field ≥ 0.5 (5×5 kinematic grid, 5 mm reach)
G4	rake	$v \cdot \hat{t} \geq -0.02$ m s^{-1} (no up-rake catch)

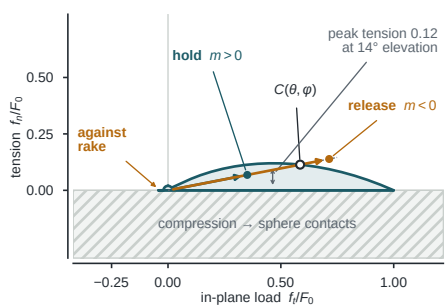
INSIDE LATCHED — age since seat



B



C



D

